Implicit Learning Dynamics in Stackelberg Games Jin, C., Netrapalli, P., and Jordan, M. I. What is local opti- mality 1
in nonconvex-nonconcave minimax optimization? arXiv preprint arXiv:1902.00618, 2019. Kelley, J. General
Topology. Van Nostrand Reinhold Com- pany, 1955. Kingma, D. P. and Ba, J. Adam: A method for stochastic
optimization. In International Conference on Learning Representations, 2015. Knutson, A. and Tao, T.
Honeycombs and sums of hermitian matrices. Notices of the American Mathematical Society, 2001. Kushner, H. J.
and Yin, G. G. Stochastic approximation and recursive algorithms and applications, volume 35. Springer Science
& Business Media, 2003. Lee, J. Introduction to smooth manifolds. Springer, 2012. Lee, J. D., Simchowitz, M.,
Jordan, M. I., and Recht, B. Gradient descent only converges to minimizers. In Con- ference on Learning Theory,
pp. 1246–1257, 2016. Letcher, A., Foerster, J., Balduzzi, D., Rockt ̈aschel, T., and Whiteson, S. Stable opponent
shaping in differentiable games. In International Conference on Learning Repre- sentations, 2019. Maclaurin, D.,
Duvenaud, D., and Adams, R. Gradient- based hyperparameter optimization through reversible learning. In
International Conference on Machine Learn- ing, pp. 2113–2122, 2015. Madry, A., Makelov, A., Schmidt, L., Tsipras,
D., and Vladu, A. Towards deep learning models resistant to adversarial attacks. In International Conference on
Learn- ing Representations, 2018. Mazumdar, E. and Ratliff, L. J. Local nash equilibria are isolated, strict local nash
equilibria in 'almost all' zero- sum continuous games. In IEEE Conference on Decision and Control, 2019.
Mazumdar, E., Jordan, M., and Sastry, S. S. On finding local nash equilibria (and only local nash equilibria) in
zero-sum games. arxiv:1901.00838, 2019. Mazumdar, E., Ratliff, L. J., and Sastry, S. S. On gradient- based learning
in continuous games. SIAM Journal on Mathematics of Data Science, 2(1):103–131, 2020. Mertikopoulos, P.,
Lecouat, B., Zenati, H., Foo, C.-S., Chan- drasekhar, V., and Piliouras, G. Optimistic mirror descent in saddle-point
problems: Going the extra (gradient) mile. In International Conference on Learning Representations, 2019.
Mescheder, L., Nowozin, S., and Geiger, A. The numerics of gans. In Advances in Neural Information Processing
Systems, pp. 1825–1835, 2017. Metz, L., Poole, B., Pfau, D., and Sohl-Dickstein, J. Un- rolled generative adversarial
networks. In International Conference on Learning Representations, 2017. Nagarajan, V. and Kolter, J. Z. Gradient
descent gan opti- mization is locally stable. In Advances in Neural Infor- mation Processing Systems, pp.
5585–5595, 2017. Ortega, J. M. and Rheinboldt, W. C. Iterative Solutions to Nonlinear Equations in Several
Variables. Academic Press, 1970. Panageas, I. and Piliouras, G. Gradient Descent Only Con- verges to Minimizers:
Non-Isolated Critical Points and Invariant Regions. In Innovations in Theoretical Com- puter Science, 2017.
Pemantle, R. Nonconvergence to unstable points in urn mod- els and stochastic approximations. Annuals
Probability, 18(2):698–712, 04 1990. Radford, A., Metz, L., and Chintala, S. Unsupervised rep- resentation learning
with deep convolutional generative adversarial networks. arXiv preprint arXiv:1511.06434, 2015. Ratliff, L. J.,
Burden, S. A., and Sastry, S. S. Genericity and structural stability of non-degenerate differential nash equilibria. In
American Control Conference, pp. 3990– 3995. IEEE, 2014. Ratliff, L. J., Burden, S. A., and Sastry, S. S. On the
Charac- terization of Local Nash Equilibria in Continuous Games. IEEE Transactions on Automatic Control,
61(8):2301– 2307, 2016. Sastry, S. S. Nonlinear Systems Theory. Springer, 1999. Sch ̈afer, F. and Anandkumar, A.
Competitive gradient de- scent. In Advances in Neural Information Processing Systems, pp. 7623–7633, 2019.
Shub, M. Global Stability of Dynamical Systems. Springer- Verlag, 1978. Thoppe, G. and Borkar, V. A concentration
bound for stochastic approximation via alekseev's formula. Stochas- tic Systems, 9(1):1–26, 2019. Tretter, C.
Spectral Theory of Block Operator Matrices and Applications. Imperial College Press, 2008. Von Stackelberg, H.
Market structure and equilibrium. Springer Science & Business Media, 2010.